

A Hybrid Machine Learning Framework for QR Code Phishing (Quishing) Detection

Yash Talaviya, Tirth Bhut, Pratham Gajra, and Prof. Yogeshwar Prajapati

Department of Computer Engineering, Marwadi University, Rajkot, Gujarat, India

Abstract— Quick Response (QR) codes are widely used for digital payments, authentication, ticketing, marketing, and information sharing because they provide a fast and convenient way to access online resources. However, their growing popularity has also led to an increase in QR code phishing (Quishing) attacks, where attackers embed malicious URLs inside QR codes to redirect users to fake websites designed to steal personal or financial information. Since users cannot identify the embedded content by simply looking at a QR code, they often scan it without verification, making such attacks difficult to detect. This paper presents a hybrid machine learning framework for detecting malicious QR codes by combining image-based and URL-based analysis. The proposed approach extracts a 21-dimensional feature vector consisting of 14 visual features from QR code images and 7 lexical features from the decoded URLs. A balanced dataset of 200,000 QR codes, including 100,000 legitimate and 100,000 malicious samples, is used for training and evaluation. Six machine learning models Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF), XGBoost, LightGBM, and a Voting Ensemble are compared to identify the most effective classifier. Experimental results show that XGBoost and LightGBM achieve the best performance, with an accuracy of 97.18% and an F1-score of 97.16%. The framework also includes a three-level risk assessment (Low, Medium, and High) to warn users before opening suspicious links, making it suitable for real-time QR phishing detection.

I. INTRODUCTION

Quick Response (QR) codes have become an important part of everyday digital interactions by providing a simple and efficient way to connect physical objects with online services. Today, they are commonly used for mobile payments, contactless restaurant menus, event ticketing, product promotions, and multi-factor authentication (MFA). Their convenience, low cost, and ease of use have made them popular across both personal and business applications. However, the increasing dependence on QR codes has also created new cybersecurity challenges, particularly the rise of QR code phishing attacks, widely known as Quishing.

Unlike traditional phishing, where attackers send malicious links directly through emails or messages, Quishing hides these links inside QR code images. Because the actual URL is encoded within the image, users cannot determine the destination before scanning it with a mobile device. In many cases, conventional email security systems focus on detecting suspicious text, harmful links, or known malicious domains, while QR code images receive little or no inspection. As a

result, phishing emails containing QR codes can bypass existing security filters and reach users without raising immediate suspicion. Once scanned, the QR code may redirect victims to fraudulent websites that imitate trusted services and attempt to steal sensitive information such as login credentials, banking details. This growing threat highlights the need for intelligent detection techniques capable of identifying malicious QR codes before users interact with them.

II. RELATED WORK

As the adoption of QR codes continues to increase across digital payments, authentication, ticketing, and marketing, researchers have developed various techniques to detect QR code phishing (Quishing) attacks. Early detection methods primarily relied on blacklist services and rule-based security systems that compared scanned URLs against databases of known malicious websites. While these approaches are effective for detecting previously identified threats, they perform poorly against newly registered phishing domains, shortened URLs, and rapidly changing attack patterns. Since cybercriminals can continuously create new domains and modify their phishing strategies, blacklist-based detection alone is no longer adequate.

To overcome these limitations, many researchers introduced machine learning techniques that analyze the URL extracted from a scanned QR code. Features such as URL length, domain length, number of subdomains, numerical digits, and protocol type are commonly used with classifiers including Support Vector Machine (SVM), Random Forest (RF), and Naïve Bayes. However, these methods depend entirely on successful QR code decoding, making them ineffective when QR codes are damaged, blurred, or intentionally altered.

Another line of research focuses on image-based analysis, where visual patterns, texture features, and deep learning models are used to identify suspicious QR codes. Although these approaches eliminate the dependency on URL decoding, they require higher computational resources and often fail to differentiate visually similar QR codes containing different URLs.

Recent hybrid approaches combine visual QR features with URL characteristics to improve detection performance. However, most existing systems rely on static analysis, use limited datasets, or analyze only one type of feature. This work addresses these limitations by proposing a lightweight hybrid framework with a 21-dimensional feature set and multi-level risk assessment for accurate real-time Quishing detection.

TABLE I LITERATURE REVIEW

Ref	Author & Year	Detection Technique	QR Code Analysis Method	Features Used	AI/ML Model	Dataset Source	Research Gap
[1]	Yalda et al., 2026	Blockchain authentication	Cryptographic chaining	Cryptographic hash, metadata, QR ID	Cryptographic	Custom Simulated Blockchain	Traditional systems lack real-time revocation and have shared vulnerabilities.
[2]	Rafsanjani et al., 2026	Static feature classification	URL lexical & host analysis	Blacklist, lexical, host, content features	Heuristic rule-based	URLhaus, PhishTank	ML models are data-dependent; excessive app permissions compromise privacy.
[3]	Nejati et al., 2026	Hybrid (Image & URL)	CNN & Lexical analysis	Spatial patterns, URL tokens	CNN, LLMs, RF, XGBoost	PhishStorm, MalwareBazaar, PhishTank	Lack of comprehensive multi-format datasets covering diverse attachment types.
[4]	OzuberK & Cinar, 2026	Image-based	CNN & ML on QR images	Grayscale histogram, LBP, GLCM	DenseNet121, LightGBM, RF	Mendeley Data (1000 images)	Need for URL-independent threat detection combined with explainable AI.
[5]	Trad & Chehab, 2026	Image-based (QR-centric)	Structural & pixel analysis	Raw QR code pixels	XGBoost, LightGBM, RF, DT	Derived from PhishStorm	URL extraction inherently exposes users to malicious payloads.
[6]	Akram et al., 2026	Structural feature analysis	Binary grid recovery	Finder, alignment, separators, timing	Pre-trained classification model	Custom synthetic dataset	Existing visual/URL methods fail on QR codes.
[7]	Akram et al., 2025	Structural feature analysis	Protocol-level & statistical	Version, ECC, module count, density	Random Forest, XGBoost	Custom Feat-DataSet-1 & 2	Deep learning models act as black boxes lacking interpretability and transparency.
[8]	Ahmed et al., 2025	Systematic review of URL-based, image-based	URL lexical	URL data tokens	Pre-trained classification model	URL Dataset	Disjointed paradigms; lack of scalable, user-friendly security solutions.
[9]	Nigam & Bhandari, 2025	URL Based Analysis	Lexical analysis	Cryptographic methods	XGBoost	Kaggle	Need for robust systems combining human awareness with technical.
[10]	Bhumika K S et al., 2025	Hybrid (ML + Heuristic)	Decoded text analysis	TF-IDF, URL length, suspicious keywords	XGBoost + Heuristic Engine	Not Reported	Consumer apps decode QR contents without verifying URL properties or parameters.

III. PROPOSED METHODOLOGY

The proposed QR Code Phishing Detection System follows a sequential processing pipeline that converts a QR code image into a phishing risk prediction. The framework consists of five major stages: dataset preparation, preprocessing and QR decoding, hybrid feature extraction, model training, and risk assessment. Figure 1 illustrates the overall workflow of the proposed system.

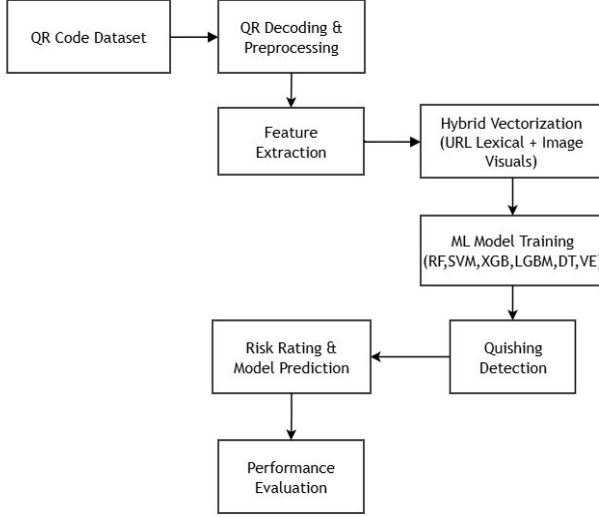


Fig. 1 Proposed Methodology

A. Dataset Preparation

The experiments are carried out using a balanced QR code dataset collected from Kaggle. The dataset contains 200,000 PNG images, equally divided into 100,000 legitimate and 100,000 phishing QR codes. Legitimate samples represent trusted websites, while phishing samples contain malicious URLs designed for fraudulent activities. Maintaining an equal number of samples in both classes helps reduce bias during model training and removes the need for oversampling methods.

B. Image Preprocessing and QR Decoding

Before feature extraction, every QR code image passes through a preprocessing stage to ensure consistent input quality and improve computational efficiency.

The image is first loaded in grayscale using the OpenCV `cv2.IMREAD_GRAYSCALE` function, which reduces memory usage and simplifies image processing. Each image is then resized to 100×100 pixels, and pixel values are normalized according to

$$I_{\text{norm}} = \frac{I(x, y)}{255.0}$$

where $I(x, y)$ represents the grayscale intensity at pixel location (x, y) .

After preprocessing, the OpenCV `QRCodeDetector` is used to decode the embedded QR content. The decoded text is cleaned using a regular-expression filter,

$$\text{Clean URL} = \text{regex_filter}(\text{Decoded Text})$$

which removes unwanted characters and extracts the valid URL. If the QR code cannot be decoded successfully, default values are assigned to the corresponding lexical features to maintain a consistent feature vector.

C. Hybrid Feature Extraction

The proposed framework generates a 21-dimensional feature vector by combining information obtained from the QR image and the decoded URL.

The 14 visual features describe the structural properties of the QR code, including an 8-bin grayscale histogram, average pixel intensity, standard deviation, image entropy, edge density, contour count, and nested contour information. These features capture the overall appearance, texture, and structural complexity of the QR code.

The remaining 7 lexical features are extracted from the decoded URL. These include URL length, domain length, number of dots, digit count, protocol type (HTTP/HTTPS), count of special characters, and a decoding status flag. Together, these features represent the characteristics commonly observed in phishing URLs.

D. Feature Selection and Data Caching

Feature extraction from thousands of QR code images is computationally intensive. To reduce repeated processing time, all extracted features are stored in a CSV file after the first execution. During subsequent experiments, the system loads these cached features directly, significantly reducing preprocessing time.

To identify the most informative attributes, ANOVA F-Classif is used for feature selection. In addition, a correlation analysis is performed to reduce features and improve the overall learning performance of the classification models.

E. Classification Models

Six machine learning models are evaluated in this study: Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), XGBoost, LightGBM, and a Voting Ensemble. The ensemble model combines the probability scores produced by Random Forest, XGBoost, and LightGBM using a soft-voting strategy, allowing the final prediction to benefit from the strengths of multiple classifiers.

$$P(y = c) = \frac{1}{3} [P_{\text{RF}}(y = c) + P_{\text{XGB}}(y = c) + P_{\text{LGB}}(y = c)]$$

The ensemble improves overall robustness by combining the strengths of multiple classifiers.

F. Risk Assessment

To make the prediction easier for users to understand, the system categorizes each scanned QR code into one of three risk levels: Low, Medium, or High. The risk level is determined using the predicted class along with the model's confidence score. High-confidence phishing predictions are labeled as High Risk, confident legitimate predictions are marked as Low Risk, and uncertain predictions are classified as Medium Risk, providing users with a clear indication of potential security threats before opening the scanned link.

IV. EXPERIMENTAL SETUP

The proposed QR code phishing detection system was implemented in **Python** using **OpenCV (4.10+)** for image processing, **Scikit-learn (1.5+)** for machine learning, and the **XGBoost** and **LightGBM** libraries for boosting models.

Experiments were performed on a system with an **Intel Core i5/i7 processor, 8 GB RAM**, and a Windows/Linux operating system. To improve processing speed, QR code images were processed in parallel using **ThreadPoolExecutor** with **16 worker threads**.

The extracted dataset of 200,000 QR codes was divided into 70% training (140,000 samples) and 30% testing (60,000 samples) using stratified sampling to maintain class balance.

Performance Metrics: The proposed models were evaluated using standard classification metrics, including Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Training Time.

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

These metrics provide a comprehensive evaluation of both detection performance and computational efficiency.

V. RESULT AND DISCUSSION

A. Performance of Individual Models

The confusion matrix was used to evaluate the classification performance of all six machine learning models. It provides a detailed view of correctly and incorrectly classified QR codes by reporting the number of **True Negatives (TN)**, **False Positives (FP)**, **False Negatives (FN)**, and **True Positives (TP)**. In phishing detection, reducing **False Negatives** is essential because undetected malicious QR codes may expose users to phishing attacks, while minimizing **False Positives** helps prevent legitimate QR codes from being incorrectly flagged.

The results were obtained using the **60,000-sample test set**, which consists of **30,000 legitimate** and **30,000 phishing QR codes**.

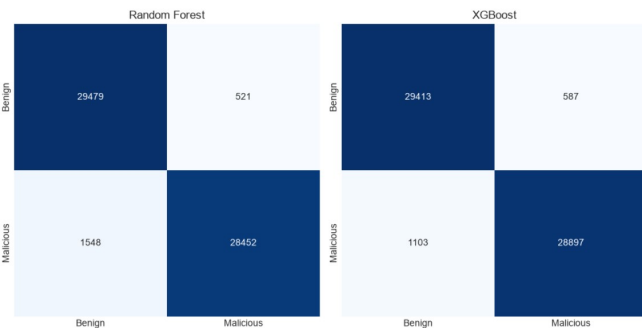


Fig. 2 Confusion matrices for RF and XGBoost

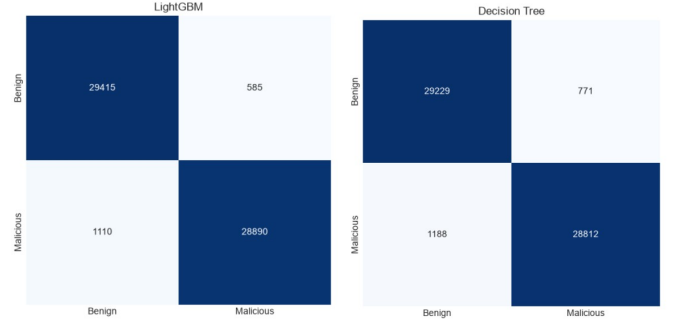


Fig. 3 Confusion matrices for LightGBM and DT

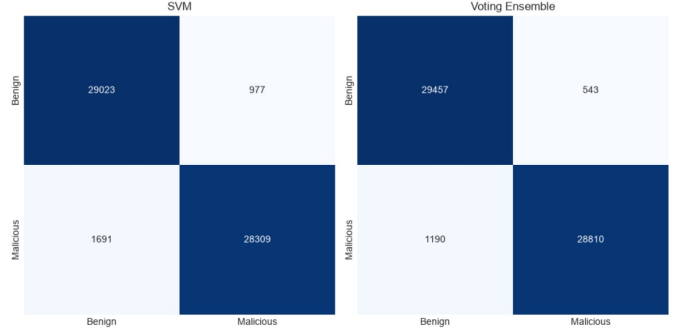


Fig. 4 Confusion matrices for SVM and VE

B. Model Result (ROC Curve)

The **Receiver Operating Characteristic (ROC)** curve was used to measure the classification ability of each model at different decision thresholds. The corresponding **ROC-AUC** value represents how well a model separates phishing QR codes from legitimate ones.

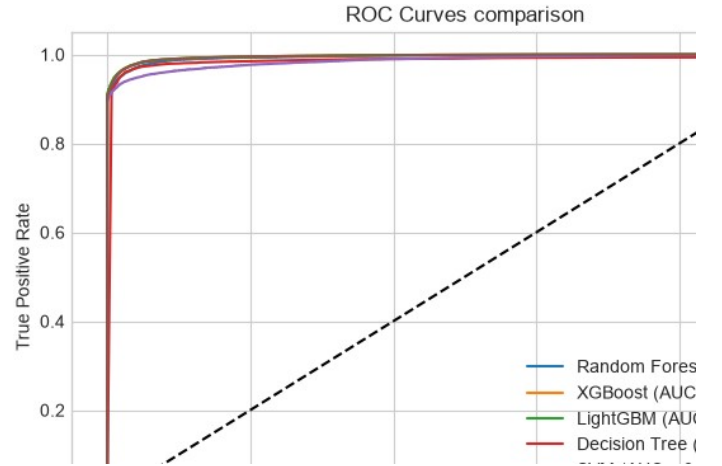


Fig. 5 ROC Curve Comparison

Among all evaluated models, **XGBoost** and **LightGBM** achieved the highest **ROC-AUC score of 0.9960**, indicating excellent discrimination between legitimate and phishing QR codes across different classification thresholds.

C. Comparative Analysis

Table II summarizes the overall performance of the six machine learning models using the selected evaluation metrics.

TABLE II COMPARISON OF MODEL PERFORMANNCE

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.9655	0.9820	0.9484	0.9649	0.9947
XGBoost	0.9718	0.9801	0.9632	0.9716	0.9960
LightGBM	0.9718	0.9802	0.9630	0.9715	0.9960
Decision Tree	0.9674	0.9739	0.9604	0.9671	0.9852
SVM	0.9555	0.9666	0.9436	0.9550	0.9864
Voting Ensemble	0.9711	0.9815	0.9603	0.9708	0.9959

D. Model Result (Testing)

The trained model was further validated using two representative QR code samples from the test dataset to verify its real-world performance.



Fig. 6 Test Instance 1: Benign



Fig. 7 Test Instance 2: Malicious

The testing results confirm that the proposed framework can successfully decode QR codes, extract both visual and URL-based features, and accurately classify them. The generated confidence score and risk level provide users with a clear indication of the security status of the scanned QR code before opening the associated link.

E. Discussion

The experimental results demonstrate that combining visual QR code features with lexical URL features improves phishing detection compared with approaches that rely on only

one type of information. Visual features such as edge density, contour patterns, grayscale statistics, and image structure help identify changes or irregularities in QR codes, while URL-based features detect common phishing characteristics such as suspicious domains, insecure protocols, and unusual URL patterns. The combination of these complementary features contributes to more reliable classification.

Among the evaluated models, **XGBoost** and **LightGBM** achieved the highest classification accuracy of **97.18%**. Both models showed excellent predictive performance, with XGBoost obtaining a slightly higher **F1-Score (97.16%)** due to its marginally better recall. These results demonstrate that gradient boosting techniques are highly effective for QR code phishing detection using the proposed hybrid feature set.

In addition, the use of parallel feature extraction and feature caching reduces preprocessing time, making the proposed framework practical for large datasets and real-time QR code phishing detection.

VI. CONCLUSION

This paper presented a hybrid machine learning framework for detecting QR code phishing (Quishing) attacks by combining visual information from QR code images with lexical features extracted from decoded URLs. The proposed approach uses a 21-dimensional feature vector consisting of 14 visual and 7 URL-based features, enabling the system to analyze both the structural characteristics of QR codes and the properties of the embedded links.

The framework was evaluated on a balanced dataset of 200,000 QR code images containing equal numbers of legitimate and phishing samples. Six machine learning classifiers were trained and compared to assess their effectiveness. Among them, **XGBoost** and **LightGBM** achieved the highest classification accuracy of **97.18%**, with XGBoost recording the best F1-Score of **97.16%**. The experimental results demonstrate that combining visual and lexical features provides better detection performance than methods that rely only on image analysis or URL analysis.

Overall, the proposed framework offers an accurate, scalable, and practical solution for real-time QR code phishing detection. The inclusion of a multi-level risk assessment mechanism further improves usability by providing clear security warnings before users access potentially malicious websites. These findings indicate that the proposed approach can effectively enhance QR code security in modern mobile and web environments.

VII. FUTURE WORK

Future research can focus on extending the proposed framework in several directions to improve its effectiveness and practical applicability. Lightweight deep learning models, such as MobileNetV4, can be explored to automatically learn visual features from QR code images while maintaining efficient performance on mobile devices. Incorporating webpage content analysis, domain reputation, and SSL

certificate verification can further strengthen the detection of newly emerging phishing websites.

Another important direction is to improve the interpretability of the model by integrating Explainable AI (XAI) techniques such as SHAP or LIME, allowing users to better understand the factors that influence each prediction. In addition, the system will be evaluated against physically modified QR codes, including damaged, partially covered, or sticker-overlaid QR codes, to assess its robustness under real-world conditions.

The proposed framework can also be deployed as a mobile application or browser extension for real-time QR code scanning and risk assessment. Evaluating the system on larger and more diverse datasets collected from different environments will further improve its reliability and generalization for practical cybersecurity applications.

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Proof of Submission:-

Submitted Ideas/PoCs							
Sr. No.	Innovation ID	Innovation Title	Sector/Domain/Theme	Team & Mentor Details	Action	First Stage of Evaluation Status	Upgrade
1	IR2026-1040213	An Intelligent Multimodal System for QR Code Phishing Detection and Mitigation	Defence & Security, ICT, cyber-physical systems, Blockchain, Cognitive computing, Cloud computing, AI & ML., IoT based technologies (e.g. Security & Surveillance systems etc)	 Team Members Details	 		

Regarding New Patent Application

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